CLINICAL · MOUTH & THROAT

Carcinoma of the hypopharynx

Cancer of the hypopharynx is the quiet killer of the throat. It grows in a wide, hidden gutter beside the larynx, gives little more than a pricking on swallowing until it is large, spreads early through a rich web of lymphatics and submucosal channels, and so presents late, often with a lump in the neck. It carries the worst prognosis of the head and neck squamous cancers, and its treatment usually involves the larynx.

AT A GLANCE

- Piriform sinus, postcricoid region, posterior wall.
- Plummer-Vinson syndrome predisposes to postcricoid cancer.
- Three-quarters have nodes at presentation.
- Look for a synchronous oesophageal cancer.

1 CHAPTER ONE Overview

The gutter beside the larynx.

1.1 Subsites

- **Piriform sinus:** on each side of the larynx, from the pharyngoepiglottic fold to the oesophagus.
- **Postcricoid region:** the anterior wall behind the cricoid lamina.
- **Posterior pharyngeal wall:** from the level of the hyoid to the cricoarytenoid joints.

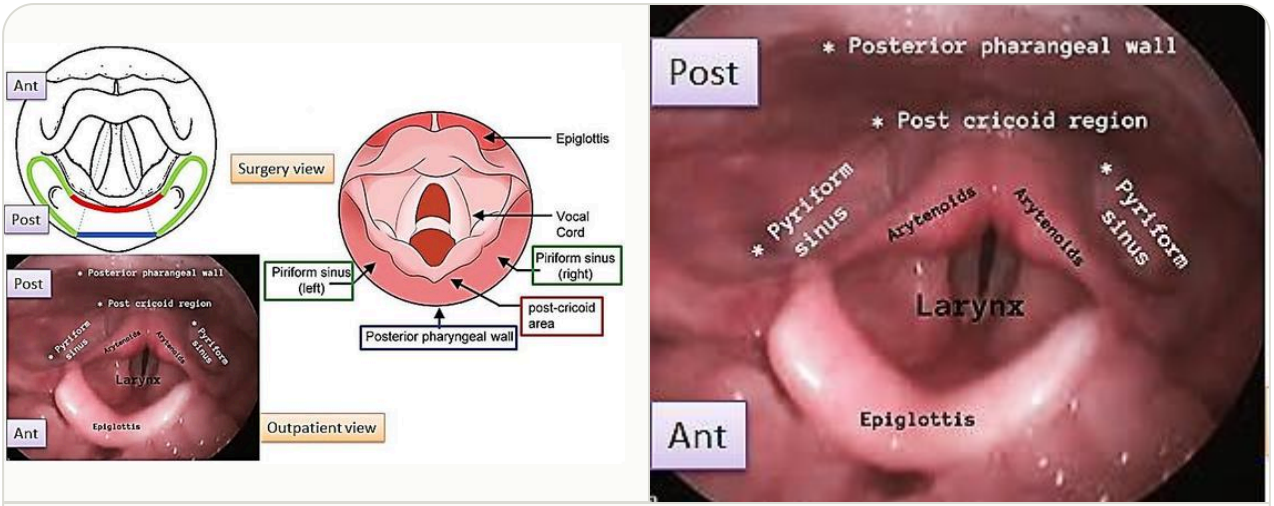


FIGURE 1 The three subsites of the hypopharynx, and an endoscopic view of the hypopharynx and larynx.

Anatomy atlas · ENT Steps p. 220, ENT Steps p. 221

1.2 Frequency and causes

Common in India. More than 90% are squamous cell carcinomas. In Dhingra's series the piriform sinus accounts for about 60%, the postcricoid region 30% and the posterior wall 10%; piriform sinus cancers predominate in North America, postcricoid cancers in parts of Europe. Tobacco and alcohol are the main causes; *Plummer–Vinson (Paterson–Brown–Kelly) syndrome* (iron deficiency anaemia, dysphagia, upper oesophageal webs, glossitis and koilonychia, in women of thirty to fifty) predisposes to postcricoid cancer.

1.3 Benign tumours

Exceptionally rare: papilloma, adenoma, lipoma, fibroma, leiomyoma, appearing as smooth, sometimes pedunculated masses.

1.4 Behaviour

- **Silent growth:** the piriform sinus is roomy, so tumours become large before they obstruct.
- **Submucosal spread:** tumours extend well beyond their visible edge, so resection margins must be generous.

- **Early lymphatic spread:** 60 to 75% have palpable nodes at presentation, often bilateral, including retropharyngeal and paratracheal nodes.
- **Second primaries:** a synchronous oesophageal cancer is found in a significant minority; always examine the oesophagus.
- **Distant metastasis** in about a fifth.

CHAPTER TWO

2 Carcinoma of the piriform sinus

The commonest.

2.1 Clinical features

Mostly men over forty. Early symptoms are few: a feeling of *something sticking in the throat* or a pricking on swallowing, localised to one side. Then pain on swallowing, referred otalgia (through the internal laryngeal and vagus nerves), progressive dysphagia, and a lump high in the neck that may be the first sign. Hoarseness and stridor indicate spread to the larynx, fixation of the hemilarynx or laryngeal oedema.

2.2 Spread

- **Up** to the vallecula and base of tongue.
- **Down** to the postcricoid region and oesophagus.
- **Medially** to the aryepiglottic fold, ventricle and paraglottic space (fixing the vocal fold).
- **Laterally** through the thyroid cartilage and thyrohyoid membrane into the thyroid gland and neck.
- **Nodes:** about three-quarters at presentation, half bilateral, to the upper and middle jugular nodes; nodes may appear long after the primary has been controlled.

POOLING OF SALIVA

On flexible laryngoscopy, saliva pooling in one piriform sinus is a sign of obstruction there; ask the patient to puff the cheeks out to open the sinus and show the tumour.

2.3 Treatment

Early tumours without nodes: radiotherapy (or chemoradiation), preserving the larynx, or selected transoral laser or robotic resection, or Ogura's extended partial laryngopharyngectomy, with neck treatment. Tumours confined to the piriform sinus but advanced: total laryngectomy with partial pharyngectomy, the remaining pharynx closed primarily, with neck dissection. Extension to the postcricoid region: total laryngopharyngectomy with neck dissection and reconstruction. Postoperative radiotherapy routinely, or chemoradiation for positive margins or extranodal spread. Many now treat advanced disease with concurrent chemoradiation, or with induction chemotherapy followed by chemoradiation in responders, to try to save the larynx, keeping surgery for salvage.

3 CHAPTER THREE Carcinoma of the postcricoid region

The woman's cancer.

3.1 Clinical features

Often women, sometimes young (in their twenties and thirties) with Plummer–Vinson syndrome, in about a third of cases. Progressive dysphagia, first for solids, with weight loss and malnutrition; voice change or aphonia when the tumour reaches the recurrent laryngeal nerve, the posterior cricoarytenoid muscle or the cricoarytenoid joint.

3.2 Spread

Circumferentially (annular), causing tight stenosis; down into the cervical oesophagus; forwards into the arytenoids and cricoid; to the paratracheal nodes, which are not palpable, often on both sides.

3.3 Signs

The tumour may be hidden on mirror examination; oedema and redness of the arytenoid region, pooling of saliva, and *loss of laryngeal crepitus* (the grating felt when

the larynx is moved across the spine) suggest it. A lateral soft tissue radiograph shows a widened prevertebral shadow.

3.4 Treatment

Prognosis is poor. Small tumours may be treated by radiotherapy or chemoradiation to preserve the larynx; most present late and need *total laryngopharyngo-oesophagectomy* with reconstruction by gastric pull-up or colon transposition, or a free jejunal graft if the oesophagus can be preserved below.

4 CHAPTER FOUR Posterior pharyngeal wall

The least common.

Mostly men over fifty. Usually exophytic, sometimes ulcerative; it stays localised for a long time, then invades the prevertebral fascia, muscles and vertebrae. Nodal spread is often bilateral and retropharyngeal; about half have nodes at presentation. Dysphagia, spitting of blood, or a neck mass without other symptoms. Early lesions are treated by radiotherapy, or by excision through a lateral pharyngotomy (or transhyoid approach, or transorally) with repair, preserving the larynx; advanced lesions by laryngopharyngectomy and neck dissection. Fixation to the prevertebral fascia makes a tumour unresectable (T4b).

5 CHAPTER FIVE Evaluation

Mapping a hidden tumour.

5.1 Examination

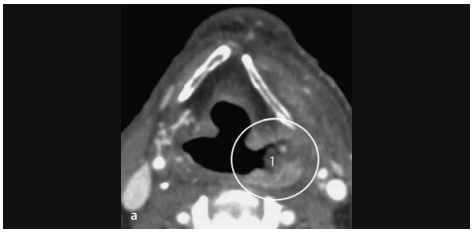
Flexible laryngoscopy of the piriform sinuses, postcricoid region and posterior wall (puffing the cheeks opens the sinuses); vocal fold mobility; palpation of the larynx for crepitus and the neck for nodes, including the thyroid and the central compartment.

5.2 Investigations

- **Barium swallow:** shows the lower limit of postcricoid and oesophageal disease, webs, and a second oesophageal tumour.
- **Contrast CT** of the neck and chest: the primary, cartilage and prevertebral invasion, and nodes; **MRI** for cartilage and soft tissue extent.
- **PET-CT:** distant metastases (lungs, liver, bone, mediastinal nodes) and second primaries.
- **Panendoscopy** under anaesthesia: biopsy, mapping of the extent (including submucosal spread) and a search for synchronous primaries, including oesophagoscopy.
- Thyroid function, haemoglobin and iron studies; nutrition and swallowing assessment.

IMAGING IN THE APP 1 case

Supraglottic and pharyngeal tumours on imaging.



Supraglottic and Pharyngeal Tumors

Tumours · 4 images

5.3 T staging

T STAGING OF HYPOPHARYNGEAL CANCER (AJCC 8TH EDITION)

CATEGORY	DEFINITION
T1	One subsite and 2 cm or less
T2	More than one subsite or an adjacent site, or over 2 cm up to 4 cm, without fixation of the hemilarynx
T3	Over 4 cm, or fixation of the hemilarynx, or extension to the oesophageal mucosa

CATEGORY	DEFINITION
T4a	Thyroid or cricoid cartilage, hyoid, thyroid gland, oesophageal muscle or central compartment soft tissue
T4b	Prevertebral fascia, encasement of the carotid artery, or mediastinal structures

6 CHAPTER SIX Treatment

Larynx and life.

6.1 Choosing

The decision balances cure against the larynx and swallowing: subsite, stage, nodal disease, the patient's fitness and wishes, and local expertise all count.

- **T1 to T2, N0:** radiotherapy (treating both sides of the neck and the retropharyngeal nodes), or larynx-preserving surgery (transoral laser or robotic resection, partial pharyngectomy, extended partial laryngopharyngectomy) with neck dissection.
- **Larger T2 and T3, or any node-positive disease:** concurrent chemoradiation; induction chemotherapy with response-guided treatment; or total laryngopharyngectomy with neck dissection and adjuvant therapy.
- **T4a:** usually total laryngopharyngectomy with neck dissection and postoperative (chemo)radiation; chemoradiation where the patient declines laryngectomy.
- **Postcricoid tumours** usually need laryngopharyngectomy, with oesophagectomy if the oesophagus is involved.

6.2 Reconstruction

- **Partial pharyngeal defects:** primary closure, or a pectoralis major, radial forearm or anterolateral thigh flap patch.
- **Circumferential defects:** a tubed fasciocutaneous free flap or a free jejunal graft.
- **With oesophagectomy:** gastric pull-up (or colon interposition).

Voice is restored by tracheoesophageal puncture and valve after laryngectomy.

OPERATIVE GUIDES IN THEATRE 5 operations

Total laryngopharyngectomy, pharyngectomy, partial pharyngectomy, reconstruction of hypopharyngeal defects and oesophagoscopy.

Total Laryngopharyngectomy
Total laryngopharyngectomy
Pharyngectomy
Partial Pharyngectomy
Partial pharyngectomy
Reconstruction of Hypopharyngeal Defects
Reconstruction of hypopharyngeal defects
Esophagoscopy and Hypopharyngoscopy
Oesophagoscopy

6.3 Outcome

Five-year survival is poor: roughly 20 to 50% for piriform sinus cancer, about a third for postcricoid cancer, and around a fifth for the posterior wall. Dhingra quotes a gross five-year cure rate of only 19% for posterior wall tumours. Distant metastases and second primaries cause many late deaths.

WHERE THE BOOKS DIFFER · WHERE THE BOOKS DIFFER

Dhingra
Advises total laryngectomy with partial pharyngectomy for piriform tumours, laryngopharyngectomy when the postcricoid region is involved, with routine postoperative radiotherapy, and radiotherapy for early tumours.

Lee
Gives a wider menu following the NCCN guidelines (induction chemotherapy, chemoradiation, transoral surgery, Ogura's partial laryngopharyngectomy) and notes that many prefer non-surgical treatment for larynx preservation.

Both agree on the late presentation, early nodal spread and poor prognosis.

7 CHAPTER SEVEN For the examination

What the examiner wants to hear.

7.1 Viva essentials

WHAT THE EXAMINER IS LOOKING FOR

- **Subsites** of the hypopharynx and their relative frequency.
- **Plummer–Vinson syndrome** and postcricoid carcinoma.
- **Why it presents late,** submucosal spread and early nodes.
- **Signs:** pooling of saliva, loss of laryngeal crepitus, hemilarynx fixation.
- **Investigations,** including barium swallow and oesophagoscopy for a second primary.
- **Treatment options** and reconstruction (jejunum, gastric pull-up, flaps).

7.2 Common pitfalls

COMMON PITFALLS

- Dismissing a localised, one-sided pricking on swallowing as globus.
- Missing a synchronous oesophageal cancer.
- Taking narrow margins in a tumour that spreads submucosally.

Written for this app from Oxford Textbook of Otolaryngology ch. 60, Oxford Specialist Handbook ch. 10, Dhingra ch. 54 and K. J. Lee ch. 45. Figures reproduced from the books for private study. Study reference only: follow local protocols and senior advice for patients.

IMAGING

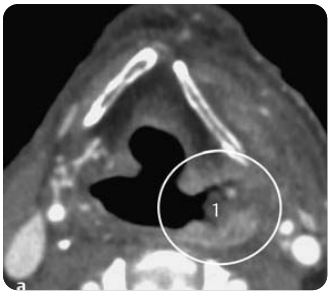
On imaging

The imaging of this topic: key findings, differentials and representative scans.

LARYNX & TRACHEA · TUMOURS

Supraglottic and Pharyngeal Tumors

- Staging is based on enlarged or pathologic-appearing lymph nodes, bilaterality, and overgrowth of the tumor on the contralateral side.
- Lymphomas (non-Hodgkin) initially might present as unilateral lymphadenopathy and ulcerative lesions.
- Evaluation of systemic disease and lymph nodes in other regions as well as secondary tumors in hypopharynx and larynx.
- Keep in mind that micrometastases may be radiologically occult.



CT with intravenous contrast enhancement, axial. Axial slice at the level of the thyroid cartilage shows a soft-tissue mass (outlined as…